24ADC001- BUSINESS INTELLIGENCE

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2026-2027
LAB RECORD

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

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Expt. No	Title of the Experiment	Page No	Total Marks	Faculty Signature
1.	Explore the interface and basic features of the BI tool			
2.	Load and visualize a sample dataset.			
3.	Import a dataset into the BI tool. And Cleanse data by handling missing values, outliers, and inconsistencies.			
4.	Transform data to suit BI reporting requirements and Design a dashboard with key performance indicators (KPIs).			
5.	Develop interactive dashboards for dynamic data exploration.			
6.	Integrate data from various sources for comprehensive analysis			
7.	Implement advanced chart types (treemaps, heatmaps, etc.).			
8.	Apply BI tools for forecasting and predictive analytics.			

Average marks **Faculty Signature**

EXPT NO:1	EXPLORE THE INTERFACE AND BASIC FEATURES OF BI Tool
DATE: 10.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1.What is the purpose of Business Intelligence (BI) tools like Tableau and Power BI?

The main purpose of BI tools such as Tableau and Power BI is to convert raw data into meaningful information that supports better decision-making. These tools help users connect to different data sources, analyse data, identify patterns and present the results through charts, reports and dashboards.

For example, a company can use Power BI to monitor monthly sales, compare the performance of different regions and identify products with declining demand. Instead of analysing thousands of rows manually, decision-makers can understand the situation quickly through visual reports.

2. List any three common data sources that Tableau and Power BI can connect to.

Three common data sources are:

1. **Excel or CSV files** – commonly used for storing small and medium-sized datasets.
2. **Relational databases** – such as MySQL, Microsoft SQL Server, Oracle and PostgreSQL.
3. **Cloud-based platforms** – such as Google BigQuery, Microsoft Azure, Amazon Redshift and Salesforce.

Both Tableau and Power BI can combine information from multiple sources to provide a complete view of business performance.

3. What is a dashboard in BI, and how is it different from a single chart?

A dashboard is a visual interface that displays multiple charts, KPIs, tables and filters in one place. It provides a combined view of important business information and helps users monitor performance at a glance.

A single chart usually explains only one aspect of the data. For example, a bar chart may display sales by region. A dashboard can display sales by region, monthly profit, top-selling products and customer growth together.

Therefore, a chart presents one specific insight, while a dashboard provides a broader and more interactive view of the business.

4(a) Define Visualization.

Visualization is the process of presenting data in a graphical form, such as charts, graphs, maps or tables. It makes complex data easier to understand and helps users quickly identify patterns, trends and unusual values.

For example, a line chart can clearly show whether a company's revenue has increased or decreased over several months.

4(b) Define KPI.

A KPI, or **Key Performance Indicator**, is a measurable value used to evaluate how effectively an organisation, department or individual is achieving an important business objective.

Examples of KPIs include total revenue, customer retention rate, profit margin, number of orders and employee productivity. KPIs help management understand whether the business is moving in the right direction.

4(c) Define Data Filter.

A data filter is a feature used to display only the data that meets a particular condition. It removes unnecessary information from the current view without permanently deleting it from the dataset.

For example, a user can apply a filter to view only sales from the year 2026, a particular region or a selected product category. Filters make reports more focused and relevant.

5. Why is it important to clean and transform data before visualization?

Cleaning and transforming data is important because inaccurate or poorly organised data can produce misleading visualizations and incorrect business decisions.

Data cleaning involves handling missing values, correcting errors, removing duplicates and standardising formats. Data transformation involves changing the structure or format of the data so that it can be analysed properly.

For example, if the same city is entered as "Chennai," "chennai" and "CHENNAI," a BI tool may treat them as separate categories. Cleaning the data ensures that all these entries are represented consistently. Reliable data leads to reliable insights.

6. What kind of chart would you use to analyse stock price trends over time?

A line chart is the most suitable chart for analysing stock price trends over time. Time is displayed on the horizontal axis, while the stock price is displayed on the vertical axis.

A line chart clearly shows increases, decreases, fluctuations and long-term trends. It also allows users to compare the prices of multiple stocks by using separate lines.

For specialised financial analysis, a candlestick chart can also be used because it displays the opening, closing, highest and lowest stock prices for a specific period.

7. Explain the role of slicers in Power BI and filters in Tableau.

Slicers in Power BI and filters in Tableau allow users to interact with a report by selecting the information they want to view.

A Power BI slicer appears directly on the report page and may allow users to select a year, region, product or customer category. When a selection is made, the related charts and tables are updated automatically.

Tableau filters perform a similar function by restricting the data shown in worksheets and dashboards. They can be applied to fields such as date, location, category or sales value.

Both features make dashboards more interactive and allow users to explore data without modifying the original dataset.

8. What does a calculated field or measure do in BI tools?

A calculated field or measure creates a new value using existing data. It allows users to perform calculations that are not directly available in the original dataset.

For example, a business may create a calculated value for profit using:

Profit = Sales Revenue − Total Cost

Other examples include profit percentage, year-over-year growth, average sales, total customers and forecast error.

In Tableau, these calculations are commonly called calculated fields. In Power BI, measures are usually created using DAX, or Data Analysis Expressions.

9. What are the advantages of using interactive dashboards for financial data analysis?

Interactive dashboards offer several advantages for financial analysis.

They provide a consolidated view of important financial indicators such as revenue, expenses, profit, cash flow and budget variance. Users can filter the dashboard by time period, department, location or product to study specific areas.

Dashboards also support drill-down analysis, allowing users to move from a high-level summary to detailed transactions. They help identify financial trends, unusual expenses and performance gaps more quickly.

Another major benefit is that dashboards can be refreshed regularly, enabling managers to make timely decisions using the latest available information.

10. Which function would you use to find the absolute difference between predicted and actual stock prices?

The ABS() function is used to calculate the absolute difference between predicted and actual stock prices.

The calculation can be written as:

Absolute Difference=ABS (Predicted Price−Actual Price)Absolute Difference=ABS(Predicted Price−Actual Price)

For example, if the predicted price is ₹150 and the actual price is ₹140:

ABS(150−140)=10ABS(150−140)=10

The result is always positive, regardless of whether the predicted value is higher or lower than the actual value. This calculation is useful for measuring prediction error in financial analysis.

IN-LAB EXERCISE:

Objective:

To understand the basic interface, data import process, and fundamental visualization capabilities of **Tableau** and **Power BI** using a stock prediction dataset, and to explore their usefulness in analyzing financial data.

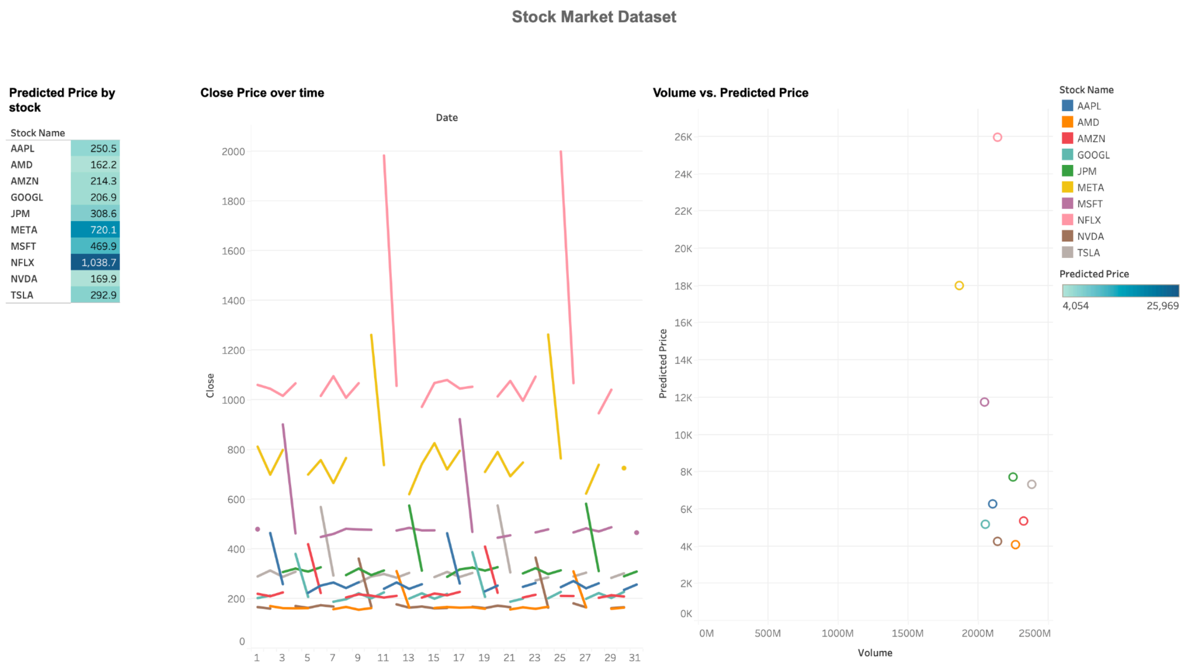
Tools Required:

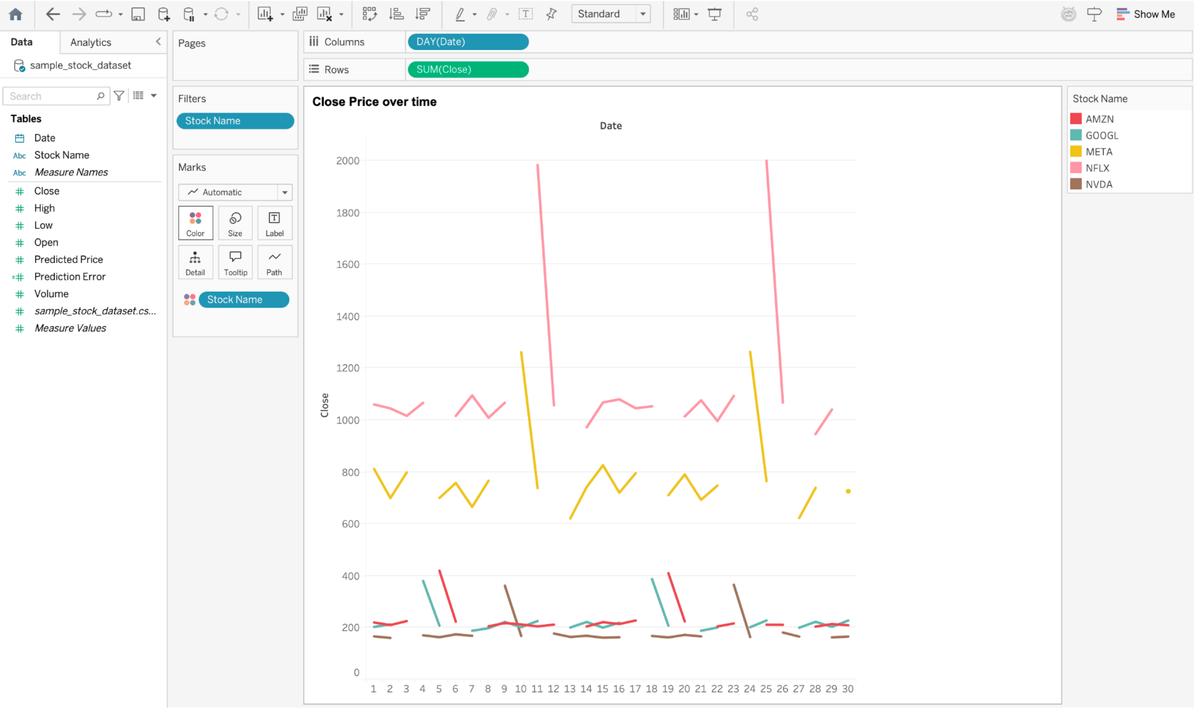
- Tableau Public / Tableau Desktop
- Power BI Desktop
- Sample Dataset (CSV/Excel) containing:
  Date, Stock Name, Open, Close, High, Low, Volume, Predicted Price

Exercise Steps:

Part 1: Working with Tableau

1. **Launch Tableau** and connect to the stock prediction dataset.
2. **Explore the Interface**: ○ Understand components like **Data Pane**, **Sheets**, **Marks Card**, and **Dashboards**.
3. **Create Basic Visualizations**:
   - ○ **Line Chart**: Show Close Price over time. ○ **Scatter Plot**: Plot Volume vs. Predicted Price.
   - ○ **Highlight Table**: Display Predicted Price by stock.
4. **Apply Filters**:
   - ○ Add filters for Stock Name and Date Range.
5. **Create Calculated Field**: ○ Calculate prediction error as |Predicted - Close|.
6. **Design a Basic Dashboard**:
   - ○ Combine your charts into one interactive dashboard.

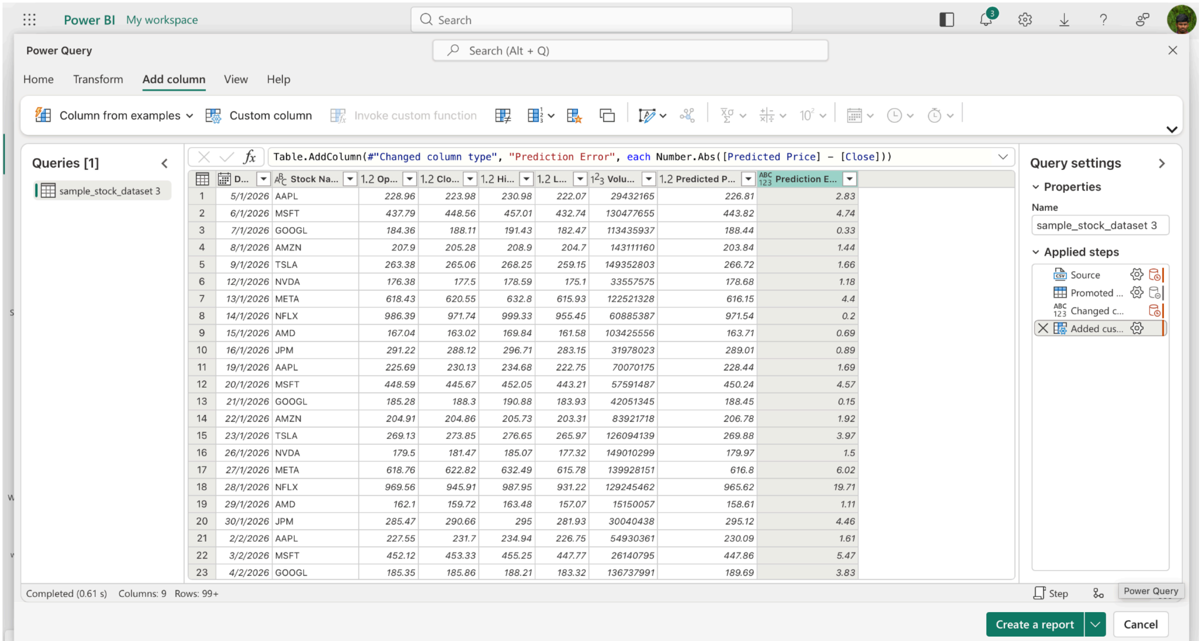


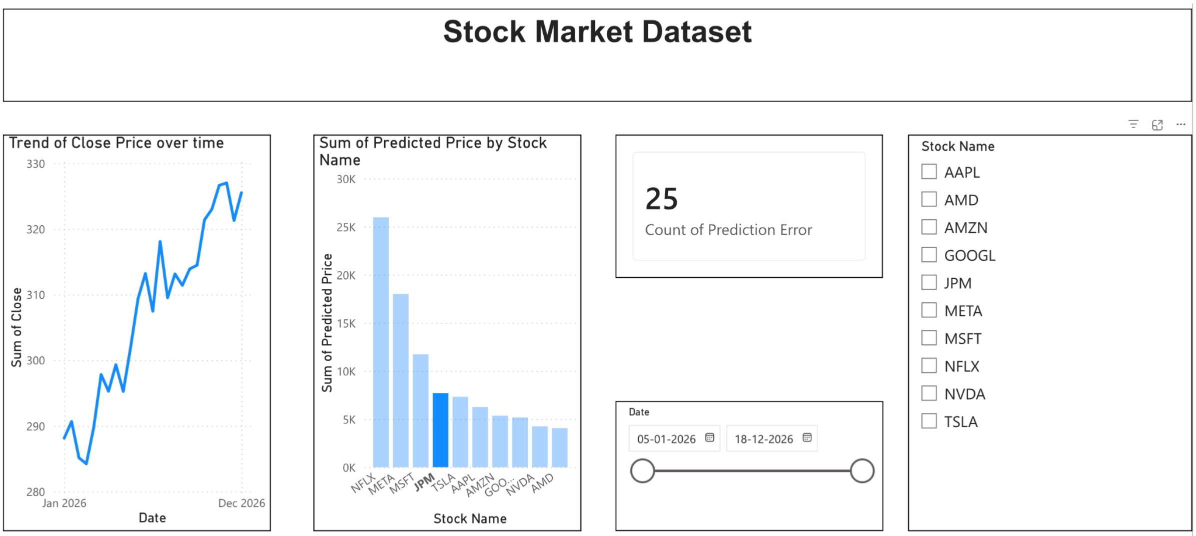


(Applying filter for line charts based on Stock Name)

Part 2: Working with Power BI

1. **Open Power BI Desktop** and import the same dataset.
2. **Explore the Interface**: ○ Familiarize with **Fields Pane**, **Visualizations Pane**, **Report View**, **Data View**.
3. **Create Basic Visualizations**:
   - **Line Chart**: Trend of Close Price over time.
   - **Bar Chart**: Compare Predicted Price vs. Close Price.
   - **Card (KPI)**: Display average prediction error.
4. **Use Filters and Slicers**:
   - Filter data based on Stock Name and Date.
5. **Create a Simple Measure** using DAX:
   - Example: Prediction Error = ABS(Predicted - Close)
6. **Build a Report Page**:
   - Arrange visuals in a single interactive page.





POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. **Describe your experience with the Tableau interface. What components did you interact with?**

My experience with Tableau was initially new, but the interface became easy to understand after some practice. I mainly worked with the Data pane, Rows and Columns shelves, Marks card, Filters shelf, Show Me panel, worksheets, and dashboard area. The drag-and-drop interface helped me build visuals quickly. I also used labels, colours, tooltips, and sorting options to make the financial data clearer and easier to interpret.

2. How did you import the dataset into Tableau and Power BI?

In Tableau, I selected the required file type from the Connect section, opened the Excel or CSV dataset, and dragged the relevant sheet into the data source area. In Power BI, I used the Get Data option, selected the file, previewed it, and clicked Load or Transform Data. In both tools, I checked that date and price columns had the correct data types before creating visualizations.

3. What types of visualizations did you create in both tools? List at least two per tool.

In Tableau, I created a line chart to compare actual and predicted stock prices over time and a bar chart to show prediction errors across different dates. In Power BI, I created a line chart for price trends and a column chart for error comparison. I also used KPI cards in Power BI to display average actual price, average predicted price, and average prediction error clearly.

4. How did you calculate the prediction error, and how was it displayed in your dashboard?

I calculated the prediction error using the absolute difference between the predicted stock price and the actual stock price. The formula was ABS(Predicted Price – Actual Price). In Tableau, I created it as a calculated field, while in Power BI I used a DAX calculated column or measure. I displayed the error using a bar chart and also showed the average error in a KPI card for quick understanding.

5. Which filtering or slicing techniques did you apply, and what insights did they reveal?

I used date filters, company filters, and prediction error ranges in Tableau. In Power BI, I used slicers for date, stock name, month, and year. These controls allowed me to focus on specific periods and companies. They revealed that prediction accuracy varied over time, with smaller errors during stable market periods and larger errors during sudden price fluctuations, helping identify when the forecasting model performed poorly.

6. Compare the process of creating a line chart in Tableau vs. Power BI. What differences did you notice?

In Tableau, I created a line chart by dragging the date field to Columns and actual and predicted prices to Rows. I then used the Marks card for colours and labels. In Power BI, I selected a line chart and placed the date on the X-axis and price measures on the Y-axis. Tableau felt more flexible for exploration, while Power BI followed a simpler, structured visual-building process.

7. How did you organize the dashboard in Tableau and Power BI? What key elements did you include?

I organized both dashboards by placing summary information at the top and detailed visualizations below. The dashboard included a title, KPI cards, a line chart comparing actual and predicted prices, a bar chart for prediction error, filters, legends, and tooltips. In Tableau, I used layout containers for alignment, while in Power BI I manually arranged and resized visuals to create a clean and balanced dashboard.

8. Which tool, Tableau or Power BI, did you find more user-friendly for financial data visualization, and why?

I found Power BI slightly more user-friendly for financial data visualization because its interface was familiar and easy to navigate. The Visualizations pane, slicers, KPI cards, Power Query, and DAX measures made it convenient to clean data and build reports in one place. Tableau offered greater flexibility and attractive visuals, but Power BI felt more suitable for creating structured financial dashboards and presenting business performance to decision-makers.

9. What challenges did you face while creating visualizations or dashboards?

Answer:
One challenge was correcting data types, especially when the date field was imported as text. I also faced difficulty placing actual and predicted prices on the same chart and writing the prediction error calculation correctly in both tools. Dashboard formatting was another challenge because visuals needed proper spacing and alignment. I solved these issues by checking field types, testing calculations, using clear labels, and maintaining a simple layout.

10. How can the knowledge gained from this lab be applied to real-world financial data analysis?

The knowledge gained from this lab can be applied to analysing stock prices, revenue, expenses, profit, budget variance, and investment performance. Financial analysts can create interactive dashboards to compare actual and forecast values, monitor KPIs, and identify unusual trends. Filters and slicers can support analysis by company, period, or department. These skills help organizations make faster, more accurate, and data-driven financial decisions.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	**5**	
In Lab Exercise	**10**	
Post Lab Exercise	**5**	
Viva	**10**	
Total	**30**	
	Faculty Signature	